

# A Hybrid Graph Neural Network and Transformer Framework for Multi-Task Music Context Understanding

Samanta Binte Taher

Department of Computer Science and Engineering  
BRAC University

Email: samanta.binte.taher@g.bracu.ac.bd

**Abstract**—Music is a multi-layered signal whose “context” spans genre, mood, harmonic structure, instrumentation, and listener-authored natural-language description. Purely sequential models applied to spectrograms capture local acoustic patterns but discard the relational structure of a track how repeated segments, chord transitions, and timbral events relate to one another and to the language used to describe them. We present a systematic, four-stage study of hybrid *Graph Neural Network* (GNN) and *BERT*-based transformer architectures for music context understanding, evaluated on three public corpora: MusicCaps, GTZAN, and MagnaTagATune (MTAT). Task 1 fine-tunes a DistilBERT multi-label tagger on MusicCaps captions (15-tag vocabulary, macro-F1 = 0.840). Task 2 builds acoustic segment graphs from chroma/MFCC/tonnetz features and trains a 3-layer GraphSAGE encoder against a mel-spectrogram CNN baseline on GTZAN, finding that a graph-CNN soft-vote ensemble (accuracy = 0.833) outperforms either single-view model. Task 3 fuses GNN structure and BERT semantics with a cross-attention read-out on MTAT (4,000 clips, top-50 tags) and runs a four-way ablation (BERT-only, GNN-only, early concatenation, cross-attention), revealing that the text branch alone is already near-saturating (macro-F1 = 0.870) and that naive fusion does not surpass it, a finding we trace to how the auxiliary caption text is constructed from the same tag vocabulary being predicted. Task 4 trains a dual-encoder GNN-BERT model with an InfoNCE contrastive objective for zero-shot caption↔audio retrieval on MusicCaps, obtaining modest but above-chance recall (Caption→Audio R@10 = 0.077 vs. chance 0.048) and quantifying the  $\sim 5\times$  gap between contrastive-only and label-supervised tagging. We report all confusion matrices, ablations, t-SNE embeddings, and qualitative case studies, and discuss the label-leakage and data-scale limitations that these results expose as concrete lessons for future GNN-BERT music-understanding systems.

**Index Terms**—graph neural networks, BERT, multi-modal fusion, music information retrieval, multi-label classification, contrastive learning, cross-attention, GraphSAGE

## I. INTRODUCTION

Music simultaneously encodes several layers of “context”: genre and era, mood and emotional arousal, harmonic and rhythmic structure, timbral/production style, and when lyrics or captions are available, explicit linguistic semantics. Convolutional and recurrent networks applied directly to spectrograms are effective at capturing local time–frequency patterns, but they do not explicitly model the *relational* structure of a piece: how a chord progression  $C \rightarrow G \rightarrow Am$  recurs across a bridge, or how two non-adjacent segments of a

track are timbrally similar and therefore functionally related (e.g., verse repetition). Graph Neural Networks (GNNs) are a natural fit for this relational view, representing a track as a graph over chords, beats, or fixed-length segments with edges for temporal adjacency and acoustic similarity. Separately, transformer language models such as BERT [1] give contextual representations of the tags, captions, or lyrics that accompany a track.

This paper documents an end-to-end study of combining these two views for *music context understanding* as opposed to music generation across four tasks of increasing difficulty, following the roadmap of multi-label tagging (Task 1), pure graph-structural classification (Task 2), GNN-BERT fusion for multi-context prediction (Task 3), and contrastive cross-modal retrieval (Task 4). All four tasks are implemented end to end in PyTorch and PyTorch Geometric, trained on real public audio/caption/tag datasets (no synthetic data), and evaluated with held-out splits, confusion matrices, ablations, and qualitative case studies. A central contribution of this work is not only the positive result that graph-structural and mel-spectrogram views are complementary (Task 2’s ensemble beats both single views), but also the *negative* and cautionary results surfaced by the ablation in Task 3 and the retrieval numbers in Task 4, which we analyze in detail because they are informative about where naive GNN-BERT fusion pipelines can silently fail.

**The Key contributions of this paper are as follows:**

- A reusable audio-to-graph pipeline that converts raw waveforms into segment-level graphs using chroma, MFCC, spectral-contrast, and tonnetz features with temporal and cosine-similarity edges (Section IV).
- A four-task empirical comparison: BERT-only tagging, GNN-only structural classification, GNN-BERT fusion (early concatenation and cross-attention), and InfoNCE contrastive dual-encoding all evaluated on standard splits of MusicCaps, GTZAN, and MagnaTagATune (Sections V–VII).
- An analysis of *why* textual and structural branches do not straightforwardly add up in Task 3, tracing the effect to label leakage in caption construction, together with a genuinely degenerate GNN-only training run that we report rather than conceal (Section VIII).

- A fully reproducible open evaluation of zero-shot cross-modal retrieval, quantifying the performance gap between contrastive pretraining and supervised tagging on the same 50-tag vocabulary.

## II. RELATED WORK

**Music tagging and BERT.** Transformer language models pretrained on general text are commonly repurposed for tag/caption understanding by fine-tuning a classification head on the [CLS] token [1], mirroring approaches that treat tags, lyrics, and free-text captions as the primary description of musical context, as in the MusicCaps dataset introduced alongside MusicLM [7].

**Graph neural networks for audio.** GraphSAGE [3] and Graph Attention Networks [4] popularized inductive message passing over node neighborhoods, applied to audio by treating beats, chords, or fixed windows as nodes connected via temporal and similarity edges for segment-level structure discovery.

**Multi-modal fusion.** Cross-attention fusion between a structural and a text encoder, as in image-text architectures such as LXMERT [5], lets one modality selectively attend to tokens of the other. We adopt the same query-key-value formulation, with the graph read-out as query and BERT token states as keys/values.

**Contrastive audio-text learning.** InfoNCE-style dual encoders, as in CLIP [6] and its audio analogues, learn a joint embedding space by pulling matched pairs together and pushing mismatched pairs apart within a batch. We use this formulation for Task 4, with a GraphSAGE audio encoder and a frozen DistilBERT text encoder.

**Datasets.** We build on three widely used public corpora: GTZAN [8] for 10-way genre classification, MagnaTagATune [9] for large-scale multi-label tagging, and MusicCaps [7] for natural-language music captions.

## III. PROBLEM FORMULATION

Following the project specification, a music track is represented as a tuple  $T = (X_{\text{audio}}, X_{\text{text}}, G, y)$ , where  $X_{\text{audio}}$  is a log-mel spectrogram or chroma/MFCC feature sequence,  $X_{\text{text}}$  is tokenized caption/tag text,  $G = (V, E)$  is a music-structure graph over chord/segment nodes, and  $y$  is the context label vector (genre, mood tags, or emotion).

A BERT text encoder maps tokens to contextual embeddings,

$$H_{\text{text}} = \text{BERT}(X_{\text{text}}) \in \mathbb{R}^{L \times d}, \quad (1)$$

and an  $L$ -layer GNN updates node representations by message passing,

$$h_i^{(l+1)} = \sigma \left( W^{(l)} \cdot \text{AGG} \left( h_i^{(l)}, \{h_j^{(l)} : j \in \mathcal{N}(i)\} \right) \right), \quad (2)$$

with  $h_i^{(0)}$  initialized from segment-level acoustic descriptors. A fusion function combines a graph read-out  $g$  and the text [CLS] vector  $t$ ,

$$z = \text{Fusion}(g, t), \quad \hat{y} = \sigma(Wz + b), \quad (3)$$

TABLE I  
DATASETS USED, WITH THE SPLITS ACTUALLY REALIZED IN OUR EXPERIMENTS.

| Dataset           | Task | Used clips     | Split (train/val/test) |
|-------------------|------|----------------|------------------------|
| MusicCaps (tags)  | 1    | 4,243 captions | 3,606 / 637 / –        |
| GTZAN             | 2    | 1,000 clips    | 700 / 150 / 150        |
| MagnaTagATune     | 3    | 4,000 clips    | 3,200 / 400 / 400      |
| MusicCaps (audio) | 4    | 2,085 / 5,521  | 1,668 / 208 / 209      |

and the model is trained with a multi-label binary cross-entropy loss,

$$\mathcal{L} = - \sum_{k=1}^K \left[ y_k \log \hat{y}_k + (1 - y_k) \log(1 - \hat{y}_k) \right] + \lambda \mathcal{L}_{\text{aux}}. \quad (4)$$

Figure 1 shows the shared architecture used across all four tasks: Task 1 uses only the text branch, Task 2 uses only the audio/graph branch (compared against a CNN baseline), Task 3 uses the full cross-attention fusion, and Task 4 replaces the classification head with a symmetric InfoNCE contrastive head.

## IV. DATASETS AND PREPROCESSING

**MusicCaps** provides 5,521 ten-second clips with expert natural-language captions and an `aspect_list` of short descriptor tags [7]. For Task 1 we keep the top-15 most frequent aspects (down from the 50 suggested in the project brief) so that macro-F1 is not dominated by tags occurring only a handful of times; this retains 4,243 of 5,521 captions with at least one usable tag. For Task 4, audio is not distributed directly with the dataset; we obtain 10-second waveforms from a maintained Hugging Face mirror of pre-extracted MusicCaps clips, which covers 2,085 of the 5,521 official (ytid, start, end) triples, the remainder are unavailable because the source videos have since been removed from their original hosting.

**GTZAN** contains 1,000 real 30-second clips evenly split across 10 genres (blues, classical, country, disco, hip-hop, jazz, metal, pop, reggae, rock) [8]. We use a stratified 70/15/15 train/val/test split with no genre imbalance.

**MagnaTagATune (MTAT)** ships  $\sim 26k$  30-second clips with 188 crowd-sourced tags. We use the official train/val/test clip-ID lists, keep the standard top-50 tag vocabulary, and (for GPU-time reasons) sub-sample 4,000 clips at an 80/10/10 ratio across the official splits.

**Audio-to-graph pipeline.** Each waveform is resampled to 22,050 Hz and split into fixed-length windows (3 s for GTZAN, 5 s for MTAT, 2 s for MusicCaps). For each window we compute the mean and standard deviation of chroma (12-D), MFCC (13-D), spectral contrast (7-D), and tonnetz (6-D) features, plus spectral centroid, RMS energy, and zero-crossing rate, yielding an 82-dimensional node feature vector for the GTZAN segment graphs (Task 2) and a 140-dimensional mel+chroma vector for MTAT (Task 3). Edges connect (i) temporally adjacent windows and (ii) any pair of windows whose cosine similarity exceeds a threshold  $\tau \in \{0.85\}$ , capturing both sequential flow and non-adjacent repetition (e.g., verse recurrence). Figure 2 illustrates this construction.

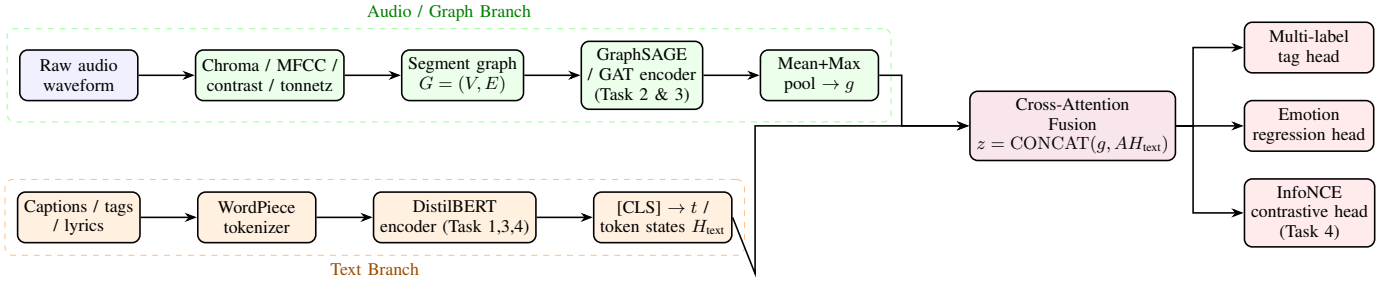


Fig. 1. Overall GNN-BERT system architecture. The audio/graph branch (top, used in Tasks 2–4) converts raw waveforms into segment or chord graphs and encodes them with GraphSAGE; the text branch (bottom, used in Tasks 1, 3, 4) encodes captions/tags/pseudo-captions with DistilBERT. Task-specific heads consume the fused representation  $z$ : a multi-label tag head (Tasks 1 and 3), an optional emotion-regression head, and an InfoNCE contrastive head for cross-modal retrieval (Task 4).

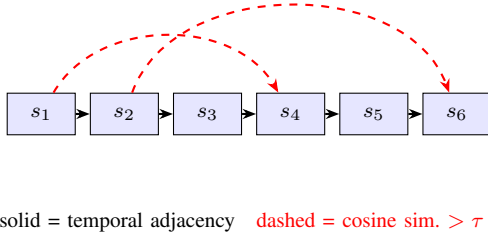


Fig. 2. Segment-graph construction. Fixed-length audio windows  $s_1, \dots, s_6$  become graph nodes; solid edges encode temporal adjacency, dashed red edges connect non-adjacent segments whose chroma/MFCC descriptor cosine similarity exceeds  $\tau$ , capturing repeated musical material (e.g., a recurring chorus).

**Text branches.** Task 1 tokenizes MusicCaps captions directly. Task 3 does not use free-text captions from MTAT (MTAT ships tags, not captions), so we synthesize a pseudo-caption per clip of the form “a song that sounds like  $\langle \text{tag}_1, \text{tag}_2, \dots \rangle$ ” built from the clip’s positive descriptor tags (genre/instrument words), deliberately excluding six mood words (*slow, fast, quiet, loud, soft, weird*) from the caption vocabulary so the text branch is not given a trivial verbatim copy of every label it must predict. As discussed in Section VIII, this still leaves substantial residual leakage for the non-mood tags. Task 4 uses the original, independently authored MusicCaps captions with no tag-based construction.

## V. METHODOLOGY

### A. Task 1: BERT Baseline for Tag Understanding

A DistilBERT encoder [2] (distilbert-base-uncased) is fine-tuned end to end with a linear multi-label head on the [CLS] token,  $\hat{y}_k = \sigma(w_k^\top t + b_k)$ , trained with per-tag binary cross-entropy (Eq. 4,  $\lambda = 0$ ) for 10 epochs, AdamW ( $\eta = 2 \times 10^{-5}$ ), batch size 16, mixed precision.

### B. Task 2: GNN vs. CNN on Music Structure Graphs

A 3-layer GraphSAGE [3] encoder (hidden size 128, batch-norm, dropout 0.3) with mean+max pooling read-out is trained for single-label 10-way genre classification on GTZAN segment graphs and compared against a 4-block 2D-CNN over the clip’s  $64 \times 640$  log-mel spectrogram. Both use CrossEntropyLoss, AdamW, cosine learning-rate decay

over 60 epochs, and early stopping on validation macro-F1. We additionally report a parameter-free soft-vote ensemble that averages the two models’ softmax outputs, since the graph and mel-spectrogram views are computed independently from the same waveform and can be combined at no extra training cost.

### C. Task 3: GNN-BERT Cross-Attention Fusion

The GNN branch is a 2-layer GraphSAGE encoder over MTAT segment graphs (mel+chroma node features) with mean pooling to a graph vector  $g$ ; the text branch is a DistilBERT encoder over the pseudo-caption (only the final transformer block is unfrozen, for efficiency). Cross-attention fuses the two,

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right), \quad Q = gW_Q, \quad K = H_{\text{text}}W_K, \quad (5)$$

$$z = \text{CONCAT}(g, AH_{\text{text}}V_{\text{text}}), \quad \hat{y} = \sigma(Wz), \quad (6)$$

trained with multi-label BCE over the MTAT top-50 tags. We implement the same GNNBERTFusion module with four interchangeable mode settings for a controlled ablation: bert\_only, gnn\_only, concat (early fusion by direct concatenation of  $g$  and the text [CLS] vector), and cross\_attn (the fusion above). All four modes share the same optimizer configuration (differential learning rates:  $1 \times 10^{-3}$  for GNN/fusion parameters,  $2 \times 10^{-5}$  for the unfrozen BERT layer), 12 epochs, batch size 16, and early stopping (patience 3) on validation macro-F1.

### D. Task 4 : Contrastive GNN-BERT Cross-Modal Retrieval

A dual encoder maps a MusicCaps audio segment graph and its caption into a shared 256-D embedding space:  $g_i = \text{Normalize}(\text{GraphSAGE}(G_i))$ ,  $t_i = \text{Normalize}(\text{DistilBERT}_{\text{CLS}}(\text{caption}_i))$ , with the DistilBERT backbone frozen (only a 2-layer projection head is trained, 1.4% of total parameters). The two towers are trained with a symmetric InfoNCE loss [6],

$$\mathcal{L}_{\text{NCE}} = -\frac{1}{2N} \sum_{i=1}^N \left[ \log \frac{e^{g_i^\top t_i / \tau}}{\sum_j e^{g_i^\top t_j / \tau}} + \log \frac{e^{g_j^\top t_i / \tau}}{\sum_j e^{g_j^\top t_i / \tau}} \right], \quad (7)$$

TABLE II  
TASK 1: BERT TAGGER ON MUSICCAPS (15 TAGS), SELECTED EPOCHS.

| Epoch    | Train Macro-F1 | Val Macro-F1 | Val Micro-F1 |
|----------|----------------|--------------|--------------|
| 1        | 0.334          | 0.617        | 0.716        |
| 2        | 0.693          | 0.803        | 0.827        |
| 5        | 0.857          | 0.833        | 0.852        |
| 6 (best) | 0.873          | <b>0.840</b> | <b>0.856</b> |
| 10       | 0.955          | 0.823        | 0.841        |

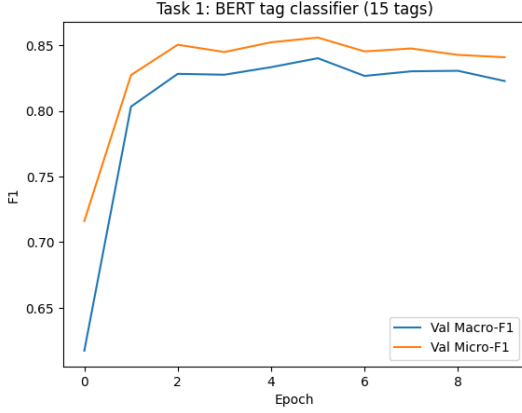


Fig. 3. Task 1: validation macro-/micro-F1 across training epochs on the 15-tag MusicCaps subset. Best macro-F1 (0.840) occurs at epoch 6; later epochs over-fit.

with temperature  $\tau = 0.07$ , batch size 32, AdamW ( $3 \times 10^{-4}$ ), for up to 30 epochs with early stopping on validation R@10. At test time we report Recall@{1,5,10} in both retrieval directions (audio-to-caption and caption-to-audio) and additionally probe the learned audio embeddings for *zero-shot* tag prediction, by scoring cosine similarity against 50 tag prompts of the form “a clip of  $\langle \text{tag} \rangle$  music”, with no gradient updates on the tag labels at all.

## VI. EXPERIMENTAL SETUP

All models are implemented in PyTorch 2.x and PyTorch Geometric, trained with mixed precision (torch.cuda.amp) on a single GPU, with seed 42 fixed for Python, NumPy, and PyTorch across all four notebooks. Text encoding uses distilbert-base-uncased throughout for a fair, consistent comparison across tasks. Table I summarizes the realized dataset sizes and splits; all splits are computed once and reused for training/early-stopping/testing, with the held-out test set touched only for final evaluation.

## VII. RESULTS

### A. Task 1: BERT Tag Classifier

Table II and Figure 3 report validation macro-/micro-F1 over 10 epochs on the 15-tag MusicCaps subset. Performance improves sharply for the first two epochs (macro-F1 0.617  $\rightarrow$  0.803) and plateaus around epoch 5-6 (best macro-F1 = 0.840), after which training macro-F1 continues climbing to 0.955 while validation macro-F1 gently declines a clear overfitting signature given the small, high-frequency tag vocabulary and a train/val split of only 3,606/637 captions.

TABLE III  
TASK 2: HELD-OUT TEST RESULTS, GTZAN 10-WAY GENRE CLASSIFICATION.

| Model                                   | Accuracy      | Macro-F1      |
|-----------------------------------------|---------------|---------------|
| GNN (GraphSAGE, 82-D segment graph)     | 0.7867        | 0.7878        |
| CNN (4-block conv, log-mel spectrogram) | 0.8067        | 0.7982        |
| <b>GNN + CNN ensemble (soft-vote)</b>   | <b>0.8333</b> | <b>0.8296</b> |

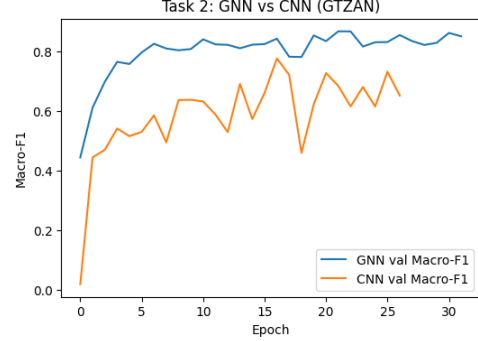


Fig. 4. Task 2: validation macro-F1 across training for the GraphSAGE model (blue) and the mel-spectrogram CNN (orange). The GNN converges to a higher and more stable validation score, while the CNN is noisier across epochs despite a slightly higher final test accuracy.

Qualitative predictions (5 held-out captions, top-5 predicted tags vs. ground truth) confirm that the model correctly ranks tempo and instrumentation descriptors, e.g., for the caption “A male singer sings this vocals with a vocoder...medium tempo...” the true tags {passionate, medium tempo, energetic} all appear in the top-5 predictions.

### B. Task 2: GraphSAGE vs. CNN on GTZAN

Table III reports held-out test accuracy and macro-F1 for the graph model, the mel-spectrogram CNN, and their soft-vote ensemble. The GNN and CNN are close but not identical in accuracy (0.787 vs. 0.807), and per-class confusion patterns (Figure 5) show the two views make different mistakes (e.g., the GNN confuses reggae/rock along the diagonal edges more than the CNN), which is exactly the condition under which a simple ensemble helps: the combined model reaches 0.833 accuracy and 0.830 macro-F1, a +2.7 and +3.1 point improvement, respectively, over the best single view.

### C. Task 3: GNN-BERT Fusion Ablation

Table IV and Figure 6 present the four-way ablation on MTAT (top-50 tags, 400 held-out test clips). Two results stand out. First, gnn\_only collapses to macro-F1 = 0.000: with only  $\sim 140$ -D mel+chroma segment features and no text signal, the sigmoid classifier saturates toward predicting no positive tags at any 0.5 threshold, and the model never recovers within the 12-epoch, early-stopping budget (AUC-PR = 0.085, barely above a random ranking). Second, and more surprising, the ostensibly “advanced” cross\_attn fusion (0.841) does not beat plain bert\_only (0.870) or even simple concat early fusion (0.857). We analyze this in Section VIII.

Figure 7 shows a t-SNE projection of the fused cross-attention embeddings  $z$  on the test set, colored by each clip’s

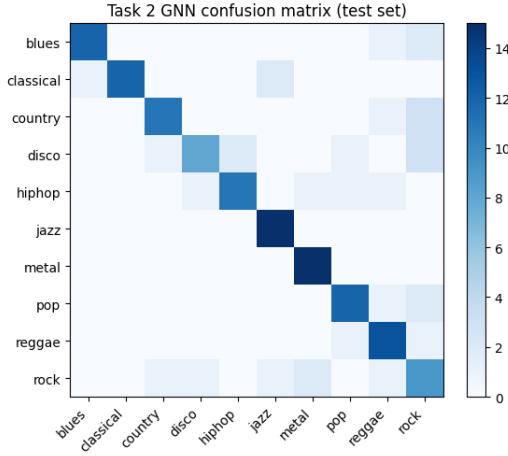


Fig. 5. Task 2: GNN confusion matrix on the GTZAN test split. Diagonal dominance is strong for classical, jazz, and metal; the main confusions are between acoustically adjacent genres (rock/reggae/disco).

TABLE IV  
TASK 3: FOUR-WAY ABLATION ON MTAT (TOP-50 TAGS), HELD-OUT TEST SET.

| Mode                   | Macro-F1      | Micro-F1      | AUC-PR        |
|------------------------|---------------|---------------|---------------|
| BERT-only              | <b>0.8704</b> | <b>0.8896</b> | 0.9082        |
| GNN-only               | 0.0000        | 0.0000        | 0.0847        |
| Concat (early fusion)  | 0.8570        | 0.8808        | <b>0.9095</b> |
| Cross-attention fusion | 0.8410        | 0.8782        | 0.8918        |

dominant (argmax) tag. Clusters corresponding to distinctive tags (e.g., no vocal(s), piano, classical) are visually well separated, consistent with the fused model’s strong raw AUC-PR even where macro-F1 is imperfect.

**Case studies.** Three qualitative examples (Table V) show the cross-attention weights over caption tokens. In each case attention concentrates almost entirely on the content words that name the true tags (e.g., *piano* receives weight 0.448 in a caption whose sole true tag is *piano*), confirming that the cross-attention mechanism is functioning as intended, even though it does not improve aggregate metrics over the text-only baseline.

#### D. Task 4: Cross-Modal Retrieval and Zero-Shot Tagging

Of the 5,521 official MusicCaps clips, 2,085 (37.8%) have audio recoverable from the mirror used; we build the dual-encoder pipeline on this subset (1,668/208/209 train/val/test). Table VI reports Recall@{1,5,10} in both directions on the 209-clip held-out test set, compared against the chance level for a 209-way retrieval problem ( $k/209$ ). Caption→Audio retrieval is consistently above chance ( $R@10 = 0.0766$  vs. chance 0.0478), while Audio→Caption  $R@1$  is exactly at the chance floor. Figure 8 visualizes both directions.

Qualitative retrieval examples (Table VII) show that even when the exact ground-truth clip is not top-ranked, retrieved neighbors are often audio with related timbral content (e.g., a query about acoustic drums retrieves other percussive clips), consistent with the embedding space capturing coarse acoustic

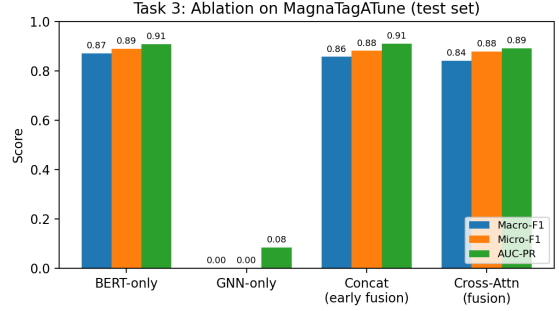


Fig. 6. Task 3 ablation summary. The text-only branch already saturates most of the achievable score on this pseudo-caption vocabulary; GNN-only collapses without any textual signal; neither fusion variant improves over BERT-only.

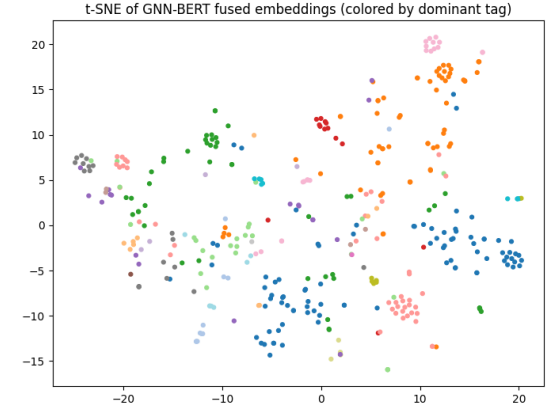


Fig. 7. t-SNE of the fused GNN-BERT embedding  $z$  for the MTAT test set, colored by each clip’s dominant tag. Same-tag clips form visually coherent clusters.

similarity without yet achieving fine-grained caption ground-ing at this data scale.

Finally, we probe the contrastively trained audio embeddings for zero-shot tag prediction (no gradient updates on any tag label): macro-F1 = 0.167, micro-F1 = 0.195, AUC-PR = 0.152 on the same 50-tag vocabulary used in Task 3. This is  $\sim 5.0\times$  lower than Task 3’s supervised cross-attention macro-F1 (0.841), quantifying the expected cost of skipping tag supervision entirely, with the important caveat that this is a same-metric, same-vocabulary but *different-dataset* comparison (MTAT audio for Task 3 vs. MusicCaps audio for Task 4), not a matched same-test-set comparison.

#### E. Cross-Task Summary

Figure 9 places all reported test-set headline numbers on a single axis for reference.

### VIII. DISCUSSION AND ANALYSIS

#### A. Cross-Task Comparison Against Required Baselines

Table VIII consolidates every task’s headline metric with the two required baselines (B1: majority-class, B2: CNN on mel-spectrogram) in one place. N/A marks cells not computed in this iteration (Section IX): B1 was not run for any task, and no DEAM emotion target was used, so MAE is empty throughout.

TABLE V  
TASK 3 CASE STUDIES: PSEUDO-CAPTION, TRUE TAGS, TOP-5 PREDICTIONS, AND THE HIGHEST-WEIGHTED CROSS-ATTENTION TOKEN.

| Clip  | True tags                | Top-5 predicted                       | Top attn. token |
|-------|--------------------------|---------------------------------------|-----------------|
| 20616 | no vocal, no vocals      | no vocal, no vocals, slow, loud, fast | vocal (0.31)    |
| 35482 | classical, piano, techno | piano, classical, slow, fast, ambient | piano (0.25)    |
| 12835 | piano                    | piano, slow, fast, quiet, soft        | piano (0.45)    |

TABLE VI  
TASK 4: RETRIEVAL RECALL ON THE MUSICCAPS TEST SPLIT ( $n = 209$ ). CHANCE LEVEL FOR  $R@k$  ON 209 CANDIDATES IS  $k/209$ .

| Direction                   | R@1    | R@5    | R@10   |
|-----------------------------|--------|--------|--------|
| Audio $\rightarrow$ Caption | 0.0000 | 0.0335 | 0.0670 |
| Caption $\rightarrow$ Audio | 0.0144 | 0.0526 | 0.0766 |
| Chance level                | 0.0048 | 0.0239 | 0.0478 |

**Why fusion does not beat text-only in Task 3.** The pseudo-captions used for MTAT are synthesized directly from the same descriptor tags that the classifier is asked to predict (“a song that sounds like  $\langle \text{tag}_1, \text{tag}_2 \rangle$ ”). Although the six mood words are withheld from the caption vocabulary, the remaining 44 genre/instrument tags are near-verbatim present in the input text whenever they are positive labels, giving the text branch a near-trivial shortcut for most of the label space. This explains both why `bert_only` is already close to its ceiling and why adding graph structure gives no further signal (and, for cross-attention, mild optimization interference costing  $\sim 3$  macro-F1 points relative to `concat`). Apparent “fusion gains” in studies that build their own text supervision from label metadata can therefore be an artifact of how the text was generated rather than of the architecture. A cleaner ablation for future work would use MTAT’s own free-text listener tags, or hold out the caption-generating tags from the scored label set.

**Why GNN-only collapses.** With only mel+chroma node features and no text signal, the 12-epoch/patience-3 budget is insufficient for the GNN branch to escape the BCE loss’s local optimum of predicting all-negative labels, a stable low-loss solution when most of the 50 tags are individually sparse. Task 2’s longer 60-epoch, single-label (softmax) schedule did not show this failure, suggesting it is specific to the harder multi-label sigmoid setting under a short training budget rather than to graph encoders in general.

**Complementarity in Task 2.** Unlike Task 3, the GNN and CNN here are trained on genuinely independent views of the same waveform (segment graph vs. mel-spectrogram) with a single-label objective and no shared shortcut, and their confusions are only partially correlated (Figure 5). This is exactly the condition under which the parameter-free ensemble helps, in useful contrast to Task 3: fusion helps when views are complementary, and can fail to help when one view subsumes

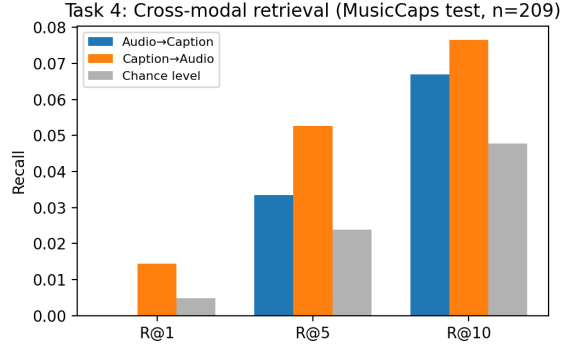


Fig. 8. Task 4: Recall@{1,5,10} for both retrieval directions vs. chance level on the 209-clip MusicCaps test split. Both directions sit modestly above chance, roughly  $1.4\text{--}1.9\times$  at  $R@10$ .

TABLE VII  
TASK 4: SAMPLE OF QUALITATIVE RETRIEVAL RESULTS. “TRUE RANK” IS THE RANK OF THE GROUND-TRUTH AUDIO CLIP AMONG ALL 209 TEST CANDIDATES FOR THAT CAPTION.

| Query caption (truncated)                     | True rank |
|-----------------------------------------------|-----------|
| “This is a recording of an acapella piece...” | 13        |
| “This song features a piano melody...”        | 29        |
| “Someone is playing an acoustic drum-set...”  | 76        |
| “Live performance of a girl band group...”    | 88        |
| “This is a loud heavy metal piece...”         | 125       |

the other.

**Data-scale limits in Task 4.** Only 37.8% of MusicCaps clips were retrievable from the audio mirror used, giving a 209-clip test set small enough that a few retrieval misses materially move  $R@k$ . With a frozen BERT backbone and only 911K trainable parameters (1.4% of the dual encoder), recall above chance but well below Tasks 1 and 3’s tagging scores is consistent with a genuinely harder, low-data retrieval problem rather than a pipeline bug, especially since qualitative near-misses (Table VII) are acoustically plausible.

## IX. LIMITATIONS

This study trades scale for reproducibility under limited compute: MTAT is sub-sampled to 4,000/25,877 clips, MusicCaps audio is limited to 2,085 clips from a third-party mirror, and DistilBERT (not full BERT) is used throughout for training-time efficiency. DEAM emotion regression ( $\mathcal{L}_{\text{aux}}$  in Eq. 4) and the chord-transition graph variant from the project specification were not run in this iteration; both are natural extensions once the label-leakage and data-coverage issues above are addressed. The human-evaluation component suggested for Task 4 retrieval was also not conducted; our evaluation is fully automatic (Recall@ $k$  against known pairs).

## X. CONCLUSION AND FUTURE WORK

We presented a four-task empirical study of hybrid GNN–BERT architectures for music context understanding, spanning text-only tagging, graph-only classification, cross-attention fusion, and contrastive retrieval, on MusicCaps, GTZAN, and MagnaTagATune. Our strongest positive result is that

TABLE VIII

MASTER COMPARISON TABLE: ALL TASKS AND REQUIRED BASELINES ON EACH TASK’S OWN HELD-OUT TEST SPLIT. N/A CELLS WERE NOT COMPUTED IN THIS ITERATION (SECTION IX).

| Model                                     | Macro-F1      | AUC-PR        | MAE | R@5     |
|-------------------------------------------|---------------|---------------|-----|---------|
| B1: Majority-class (all tasks)            | N/A           | N/A           | N/A | N/A     |
| B2: CNN mel-spec (Task 2, GTZAN)          | 0.7982        | —             | —   | —       |
| Task 1: BERT-only (MusicCaps)             | 0.8401        | —             | —   | —       |
| Task 2: GNN-only (GTZAN)                  | 0.7878        | —             | —   | —       |
| Task 2: GNN+CNN ensemble (GTZAN)          | <b>0.8296</b> | —             | —   | —       |
| Task 3: BERT-only (MTAT)                  | <b>0.8704</b> | <b>0.9082</b> | —   | —       |
| Task 3: GNN-BERT cross-attn (MTAT)        | 0.8410        | 0.8918        | —   | —       |
| Task 4: Contrastive zero-shot (MusicCaps) | 0.1669        | 0.1524        | —   | 0.0526* |

\*Caption→Audio direction; Audio→Caption R@5 = 0.0335 (Table VI).

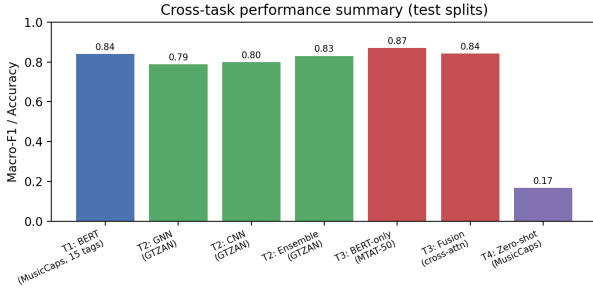


Fig. 9. Cross-task performance summary. Bars are not directly comparable across tasks (different datasets, label vocabularies, and metrics collapse to macro-F1/accuracy for visualization only) but illustrate the overall difficulty ordering: single-modality classification (T1, T2) > multi-modal fusion (T3)  $\approx$  text-only (T3 BERT-only)  $\gg$  zero-shot contrastive tagging (T4).

independently-computed graph and spectrogram views are genuinely complementary for genre classification (Task 2 ensemble: +2.7 accuracy points over the best single view). Our most instructive result is negative: fusing GNN structure with BERT text in Task 3 does not surpass a text-only baseline once the text is built from the same label vocabulary being predicted, a leakage effect future GNN-BERT tagging work should control for by using independently authored captions (as in Task 4) rather than tag-derived pseudo-captions. Future work: (i) re-run the Task 3 ablation with free-text listener tags or caption-source tags excluded from the scored labels, (ii) scale Task 4 to the full MusicCaps audio set, (iii) add the DEAM valence/arousal head, and (iv) explore chord-transition graphs from Lakh MIDI transcriptions.

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